TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	63.6 ± 3.67
Correlation Stability	89.7 ± 0.46
Distribution Consistency	67.2 ± 5.02
Entropy Preservation	47.9 ± 2.30
Mean Z Score Drift	0.3 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	30.0 ± 5.62
Data Density	100.0 ± 0.00
Coverage Score	55.7 ± 5.42
Feature Accuracy	20.0 ± 1.61
Novelty Score	0.6 ± 0.02

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Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	2.860	-0.236	7.6%	OK
PRE4	3.282	3.207	-0.075	2.3%	OK
PRE5	4.569	2.356	-2.213	48.4%	Drift
PRE6	0.781	0.845	+0.064	8.2%	OK
PRE7	0.066	0.000	-0.066	100.0%	Drift
PRE8	0.145	0.030	-0.114	79.1%	Drift
PRE9	0.066	0.004	-0.062	94.3%	Drift
PRE10	0.687	0.795	+0.108	15.7%	Warn
PRE11	0.166	0.072	-0.094	56.6%	Drift
PRE14	11.736	11.723	-0.013	0.1%	OK
PRE17	0.074	0.015	-0.059	79.7%	Drift
PRE19	0.004	0.000	-0.004	100.0%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	63.6 ± 3.67
Correlation Stability (*2)	89.7 ± 0.46
Distribution Consistency (*3)	67.2 ± 5.02
Entropy Preservation (*4)	47.9 ± 2.30
Mean Z Score Drift (*5)	0.3 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	30.0 ± 5.62

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	55.7 ± 5.42
Feature Accuracy (*10)	20.0 ± 1.61
Novelty Score (*11)	0.6 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.2606382978723404	-0.5946808510638297	*
F1	N/A	N/A	0.2281796621683534	0.2281796621683534	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	N/A	0.5670212765957447	-0.28404255319148936	*
F1	0.037675637056442016	N/A	0.18010362991917647	0.14242799286273444	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8521276595744682	N/A	0.7031914893617022	-0.14893617021276598	*
F1	N/A	N/A	0.23412428655075712	0.23412428655075712	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.5436170212765956	-0.3117021276595745	*
F1	N/A	N/A	0.23165772337586515	0.23165772337586515	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.85	N/A	0.5319148936170213	-0.3180851063829787	*
F1	0.01	N/A	0.14424731771785593	0.13424731771785595	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8127659574468085	N/A	0.6606382978723404	-0.1521276595744681	*
F1	0.11944461644306847	N/A	0.2059124849991482	0.08646786855607975	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8351063829787234	N/A	0.6851063829787234	-0.14999999999999997	*
F1	0.07477350427350428	N/A	0.17162023367905724	0.09684672940555293	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	N/A	0.225531914893617	0.0148936170212766	
F1	0.256855620661189	N/A	0.2056925661885896	-0.05116305447259937	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	N/A	0.5234042553191489	-0.3244680851063829	*
F1	N/A	N/A	0.1762273207047464	0.1762273207047464	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	N/A	0.7021276595744681	-0.11170212765957448	*
F1	0.11972345964178996	N/A	0.17790674993281372	0.05818329029102378	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.74559715707364 6	1.92803316747412 71	1.0	0.0	1.0	0.0	1.0	0.0	11.0
2	3.0	2.58423075733924 1	1.90414200451645 59	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	3.20067199498007 2	2.38016067647197 76	2.0	0.0	0.0	0.0	1.0	1.0	11.0
4	3.0	3.31524237911504 1	2.57093279455174 97	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	4.24427053536752 6	2.99500750858078 5	1.0	0.0	0.0	0.0	1.0	0.0	13.0
6	3.0	2.77467944493782 74	1.67398281586736 53	2.0	0.0	0.0	0.0	1.0	1.0	11.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.75888230540035 73	2.13944115270017 84	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.73475215299112 36	2.22025702652772 37	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.47239327913939 4	1.93423024387272 72	1.0	0.0	0.0	0.0	1.0	1.0	12.0
4	2.0	2.71303989035453 16	1.73044016446820 21	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	4.0	2.50349141877346 16	1.83622721766914 03	1.0	0.0	0.0	0.0	0.0	0.0	12.0
6	3.0	3.58175218883462 64	2.25287212371923 84	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.5314143794263186	1.7580479314210622	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.4499987346493003	2.0826663292398144	2.0	0.0	0.0	0.0	1.0	1.0	11.0
3	3.0	3.1838309839438206	2.5903815601879447	1.0	0.0	0.0	0.0	1.0	0.0	14.0
4	4.0	3.4025254576204804	2.0850509152409615	1.0	0.0	0.0	0.0	0.0	1.0	11.0
5	2.0	3.8968186860061844	3.011471566953181	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	5.0	3.2370870897880266	2.08	1.0	0.0	0.0	0.0	0.0	0.0	11.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	3.028056895186379	2.0146489003388712	1.0	0.0	1.0	0.0	0.0	0.0	13.0
2	2.0	2.9247349664345044	2.2517550111884987	0.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	3.0989566451141255	1.7431692329172643	1.0	0.0	0.0	0.0	1.0	1.0	12.0
4	3.0	2.5131641248636209	1.8429779998716507	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	2.0	2.7899120616824899	1.6383341656065415	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	2.5390454280846899	2.0998090856169308	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.195160717072369	2.381120301925208	1.0	0.0	0.0	0.0	0.0	0.0	13.0
2	3.0	3.008096734508673	2.351461236549467	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.547394522861792	1.811315076205974	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.560511702390517	1.790895479183405	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.451006703276945	2.082201340655389	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.149346906003748	1.575477524802998	1.0	0.0	0.0	0.0	1.0	0.0	13.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.655484179185582	2.742581151519171	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	2.559789557225651	1.865895043160720	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	2.810841397843186	1.675889784902302	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.813398298986473	1.919902551520289	0.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	3.010086065843232	2.022960381345992	0.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.916799517268764	1.977172303506697	2.0	0.0	0.0	0.0	1.0	1.0	11.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.1869121919997156	2.5144239840000235	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.093804650360683	1.9960930072136627	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.3134961420863482	2.157790974944192	0.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.517896786917985	1.729464458869067	2.0	0.0	0.0	0.0	1.0	1.0	12.0
5	4.0	2.4282181512745207	1.7345172928568449	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	2.0	2.907194840332511	2.4756259787110015	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.498270631702062	3.031337732420232	0.0	0.0	0.0	0.0	0.0	0.0	11.0
2	2.0	2.9466679712739636	2.2444440095753455	0.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.7698936975831856	2.3706183560989449	1.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	2.9347206336975216	2.5319088995933594	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.127100754027474	1.447992829711328	0.0	0.0	0.0	0.0	0.0	0.0	12.0
6	2.0	2.4871631074643994	1.845969256220333	0.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.49403263948575 5	1.68404519313412 2	1.0	0.0	0.0	0.0	1.0	1.0	11.0
2	2.0	3.94053356453315 6	3.02088927422192 64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	2.58901645084998 6	2.33437171864996 7	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	2.90111903402017 82	3.19724519865851 7	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	2.74291962228676 5	2.12038636301485	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.49314811663627 1	2.07269549075762 74	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.80501782436865 85	1.63512477058060 98	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.28538362249341 63	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.58225965789174 6	2.57633016666396 04	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	2.0	3.94218532930830 7	3.02364221551384 5	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	3.35055155822854	2.50078737014762 18	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	2.0	2.83121922148068 93	1.81853455036482 45	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0